Where do the other worlds go?
A toy-model test of Born-rule filtering in a local Ising chain

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Strasberg and collaborators showed, in random matrix models, that Born-atypical quantum histories recohere while Born-typical ones remain decoherent. We test this in a 9-qubit transverse-field Ising chain ($D = 512$): histories with Born-typical frequencies remain 3.1× more decoherent than those at the branch-counting peak, and individual-branch tracking confirms that survivors cluster at Born-typical frequencies. The effect requires scrambling dynamics and is robust across system sizes ($D = 128$–1024) and projector choices. When 8 qubits are split into two spatial groups with independent projectors, Born filtering operates simultaneously at each level of coarse-graining. At the branch level, coarse-grained worlds can remain decoherent even after all their fine-grained constituents have recohered.

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I. INTRODUCTION

Quantum mechanics has a puzzle at its core, and it is surprisingly easy to state.

The Schrödinger equation is deterministic and linear. If you know the state of a system right now, you can compute its state at any future time—no randomness, no choices, no collapse. And yet, every time we open the box and look at the cat, we see one definite outcome. The math says the cat is in a superposition of alive and dead. Our eyes say it is one or the other.

This is the measurement problem, and it has been with us since the 1920s. There is no shortage of proposed solutions. The Copenhagen interpretation says the wave function collapses upon measurement, but never quite explains what counts as a measurement. Bohmian mechanics adds hidden variables. Objective collapse theories add new physics. And then there is Everett's proposal [1]: nothing collapses at all. The Schrödinger equation is the whole story. What we experience as "collapse" is the universe splitting into branches—every possible outcome happens, each in its own branch of reality, and we simply find ourselves in one of them.

The Everett picture is elegant, but it raises its own question. If every outcome happens, why do we observe outcomes with the frequencies predicted by the Born rule? If there is one branch where the electron goes left and one where it goes right, and both are equally real, why do we not see 50/50 statistics regardless of the quantum state?

This is sometimes called the *probability problem* of the many-worlds interpretation, and it is the subject of a large literature. Deutsch [4] gave the first decision-theoretic argument: a rational agent in a branching universe *should* weight branches by their squared amplitudes. Wallace [2] refined and extended the proof; Carroll [3] offers an accessible account and argues, with Sebens, that Born's rule follows from self-locating uncertainty at the moment of branching. Zurek [5] pursued a different strategy, deriving Born's rule from environment-assisted invariance ("envariance") without decision theory. These approaches are persuasive to some and unconvincing to others. What they share is that they introduce additional postulates—rationality axioms or symmetry principles—beyond the Schrödinger equation itself.

Recently, Strasberg and collaborators [8–10] have explored a different route. In a finite-dimensional Hilbert space, the histories of a quantum system cannot all remain decoherent as the number of branching events grows. There are more histories than dimensions, so branches must start overlapping. In random matrix toy models, they reported that this overlap is not uniform: the histories whose frequencies match the Born rule remain decoherent, while the others do not. Whether this filtering pattern is a generic feature of finite-dimensional quantum mechanics or specific to the models studied remains an open question.

Strasberg's results are established for random matrix Hamiltonians and related asymptotic arguments. We ask: does the same qualitative behavior appear in a concrete, physically motivated Hamiltonian? We take a 9-qubit transverse-field Ising chain—a nearest-neighbor spin model near its critical point—and check.

II. THE PUZZLE: ALL BRANCHES EXIST, BUT THE STATISTICS ARE WRONG

Take a loaded coin: 75% heads, 25% tails. We know what probability means here. Flip it many times, and about 75% of the flips will come up heads. That is what we observe. That is what we mean by the coin being "loaded."

Now flip it once in the Everett picture. Both outcomes happen. There are two branches: one where you got heads, one where you got tails. Both are real. Both exist. So what does "75% probability" even mean, if both outcomes occur with certainty?

Flip again. Now there are four branches: HH, HT, TH, TT. All four exist. Of these, one has 0 heads, two have

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1 head, and one has 2 heads. If you just count branches, the most common outcome is 50% heads—not 75%.

After three flips, there are 8 branches. The branch-counting peak is still at 50% heads. After ten flips, there are 1024 branches, and $\binom{10}{5} = 252$ of them have exactly 50% heads, while only $\binom{10}{7} = 120$ have 70% heads. The more you flip, the worse it gets: branch counting says 50%, but we observe 75%.

This is the probability problem of the many-worlds interpretation. If every branch is equally real, what does probability mean? And why do we see the loaded-coin statistics instead of the branch-counting statistics?

III. BACKGROUND: RECOHERENCE IN FINITE HILBERT SPACE

A quantum system with m qubits lives in a Hilbert space of dimension $D = 2^m$. Think of this as a pond with a fixed surface area. Each branch of the wave function is a lily pad. After L coin flips, there are 2^L lily pads. When L is small, there is plenty of room—each pad has its own patch of water, and they do not interfere. But the pond is finite. When $2^L > D$, the pads are competing for the same surface. They crowd, overlap, and shade each other out. Not all can survive as distinct plants.

In quantum language, branches that "overlap" are no longer orthogonal. Their quantum states point in nearly the same direction, which means they cannot be distinguished by any measurement. They have lost their identity as separate worlds. This language draws on the decoherent (or consistent) histories framework [6, 7], in which a set of histories counts as "classical" only when their mutual overlaps are negligible. Following Strasberg [10], we call the loss of this condition *recoherence*—the opposite of decoherence.

Strasberg and collaborators [8, 10] reported that, in their random matrix models, this competition is not democratic. The branches that get shaded out are not random. They are the ones near 50% heads—the branch-counting peak. The branches near 75% heads—the Born frequency—remain more nearly decoherent.

A. Measuring Decoherence

To make this precise, we need a way to measure how much two branches overlap.

We group branches by their frequency—the total number of heads n_1 out of L flips. The frequency-class state $|\psi(n_1)\rangle$ is the quantum state obtained by adding up all branches that got exactly n_1 heads:

$$|\psi(n_1)\rangle = \sum_{\mathbf{x}:\, n_1(\mathbf{x})=n_1} \Pi_{x_L} U_L \cdots \Pi_{x_1} U_1 |\Psi_0\rangle, \qquad (1)$$

where U_k is the evolution between flips and Π_{x_k} picks out the heads-or-tails outcome at flip k.

The *recoherence parameter* ε [8] measures how much a frequency class overlaps with any other frequency class—a score from 0 to 1:

$$\varepsilon(n_1) = \max_{n_1' \neq n_1} \frac{|\langle\psi(n_1')|\psi(n_1)\rangle|}{\sqrt{\langle\psi(n_1)|\psi(n_1)\rangle \cdot \langle\psi(n_1')|\psi(n_1')\rangle}}. \qquad (2)$$

At $\varepsilon = 0$, the frequency class is fully decoherent (orthogonal to all others). At $\varepsilon = 1$, it is fully recoherent (indistinguishable from another class).

B. The Prediction

In their random matrix toy models, Strasberg and collaborators found that, as the number of flips L grows:

- Frequency classes near 75% heads have low ε—they stay decoherent,
- frequency classes near 50% heads have high ε—they blur together,
- the gap grows with L.

The pond is too small for all the lily pads near 50%. They shade each other out. The ones near 75% remain more nearly decoherent.

IV. NUMERICAL EXPERIMENT

These results are established for random matrix Hamiltonians and in asymptotic limits. Whether the same filtering pattern survives in a small, structured, physically motivated system is untested. We check.

We simulate repeated coarse-grained measurements on a 9-qubit Ising chain and analyze the decoherence structure of the resulting branches.[1]

The setup can be thought of as a loaded quantum coin flip repeated L times. The coin comes up heads 75% of the time. In the many-worlds picture, every outcome sequence occurs as a branch. We track all 2^L branches and compute the recoherence parameter ε for each, asking: which branches remain approximately decoherent, and which lose their identity through overlap with neighbors?

A. Setup: The Ising Chain

We use 9 qubits arranged in a line, interacting through the transverse-field Ising Hamiltonian:

$$H = \frac{J}{2} \sum_{i=0}^{7} Z_i Z_{i+1} + h_x \sum_{i=0}^{8} X_i, \qquad (3)$$

[1] All simulation code is available at https://github.com/eschnou/quantum-recoherence.

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with parameters $J/2 = 0.5$ and $h_x = 0.9045$, placing the system near the quantum critical point ($h_x/J \approx 0.9$). In this regime, the nearest-neighbor couplings generate significant entanglement and spreading of correlations. The Hilbert space has dimension $D = 2^9 = 512$.

Each "coin flip" is not a physical measurement but an analytical decomposition: let the qubits evolve under H for one time step, then project the state onto two orthogonal subspaces. Our question is: are at least 4 of the 9 qubits in state $|1\rangle$? If yes, we call it "heads." If no, "tails." Since 382 of the 512 basis states have Hamming weight ≥ 4, the coin is loaded: $p_{\text{heads}} = d_1/D = 382/512 \approx 75\%$, where d_1 is the rank of the "heads" projector. In Strasberg's framework, this ratio d_1/D defines the Born-typical frequency: it is the single-trial Born probability averaged over Haar-random initial states.

B. Methods

Each evolution step applies $U = e^{-iH\Delta t}$ with $\Delta t = 1.2$, computed via exact matrix exponentiation. The time step was chosen large enough for the dynamics to spread correlations across the chain between projections; smaller values ($\Delta t \lesssim 0.5$) produce weaker filtering because the state has not evolved far from the previous projection subspace.

Initial states are drawn Haar-randomly on the full D-dimensional Hilbert space. All frequency-class results are averaged over 10 independent random seeds; error bars show $\pm 1\sigma$.

Branch-level decoherence counts use the operational threshold $\varepsilon < 0.3$. This threshold is a conventional choice; the qualitative conclusion—that survivors cluster at Born-typical frequencies—holds for thresholds in the range $[0.1, 0.5]$, though the absolute count of "decoherent" branches varies (see Appendix A 3 for the full threshold sensitivity analysis). A non-scrambling control (product rotation $U_{\text{prod}} = e^{-i\theta X \otimes m}$ at matched p_1) produces no stable filtering, confirming that scrambling dynamics are required (Appendix A 4).

C. Frequency-Class Results

We first analyze frequency classes (groups of branches with the same number of heads) at $L = 15$, averaged over 10 Haar-random initial states.

Figure 1 shows the result. The frequency classes near 75% heads (the Born prediction) have low ε—they remain approximately decoherent. The classes near 50% heads (the branch-counting peak) have ε approximately three times higher—they overlap significantly with neighboring classes.

As a sanity check, we also run Strasberg's own random matrix model at the same dimension and coin loading. The two curves show the same qualitative pattern—high

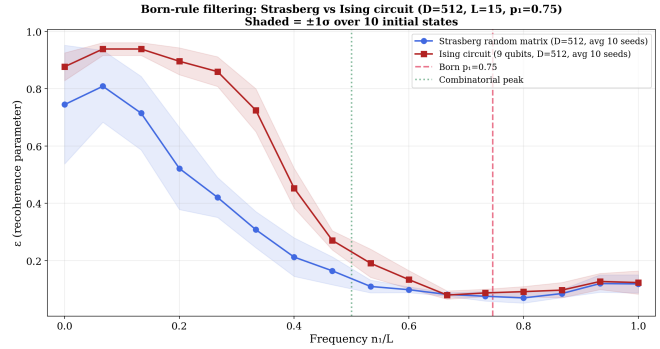

FIG. 1. Recoherence parameter $\varepsilon(n_1/L)$ for the 9-qubit transverse-field Ising chain (red, exact matrix exponentiation) and Strasberg's random matrix model (blue) at $D = 512$, $L = 15$, $p_1 = 0.746$. Averaged over 10 Haar-random initial states; shaded regions show $\pm 1\sigma$. Both curves dip near the Born frequency (red dashed) and rise toward the combinatorial center (green dotted), showing the same qualitative filtering pattern.

ε at combinatorial-peak frequencies, low ε near the Born frequency.

The result is robust. All 10 seeds show the same qualitative shape. The standard deviation of ε across seeds is small relative to the Born–combinatorial gap, indicating that the effect is a property of the Hamiltonian, not of special initial states. The effect also persists across system sizes ($D = 128$ to 1024) and alternative coarse-grainings; see Appendix A for the full scaling and projector analysis.

D. Branch-Level Analysis

So far we have grouped branches by their frequency (how many heads total). But we can go deeper: track every single branch individually. Each specific sequence of heads and tails—like HHTHTTHHTTHH—is one branch, one possible history of the coin. We compute ε for each one: how much does this branch overlap with any other?

Figure 2 shows what happens as we increase the number of flips. At $L = 4$ (16 branches, 512 dimensions), the pond is nearly empty—almost every branch is a distinct world. At $L = 9$ (512 branches, 512 dimensions), the pond is full. At $L = 12$ (4096 branches, 512 dimensions), it is eight times oversubscribed. Only 451 branches remain approximately decoherent ($\varepsilon < 0.3$). By $L = 13$ (8192 branches), 565 survive—slightly above D, since the threshold $\varepsilon < 0.3$ allows partial overlap.

The survivors cluster at Born-typical frequencies.

Figure 3 makes the census explicit. The total number of branches grows exponentially (2^L), but the growth of decoherent branches slows dramatically around $D = 512$. The dimension D is not a hard ceiling—approximate decoherence ($\varepsilon < 0.3$) allows more than D coexisting

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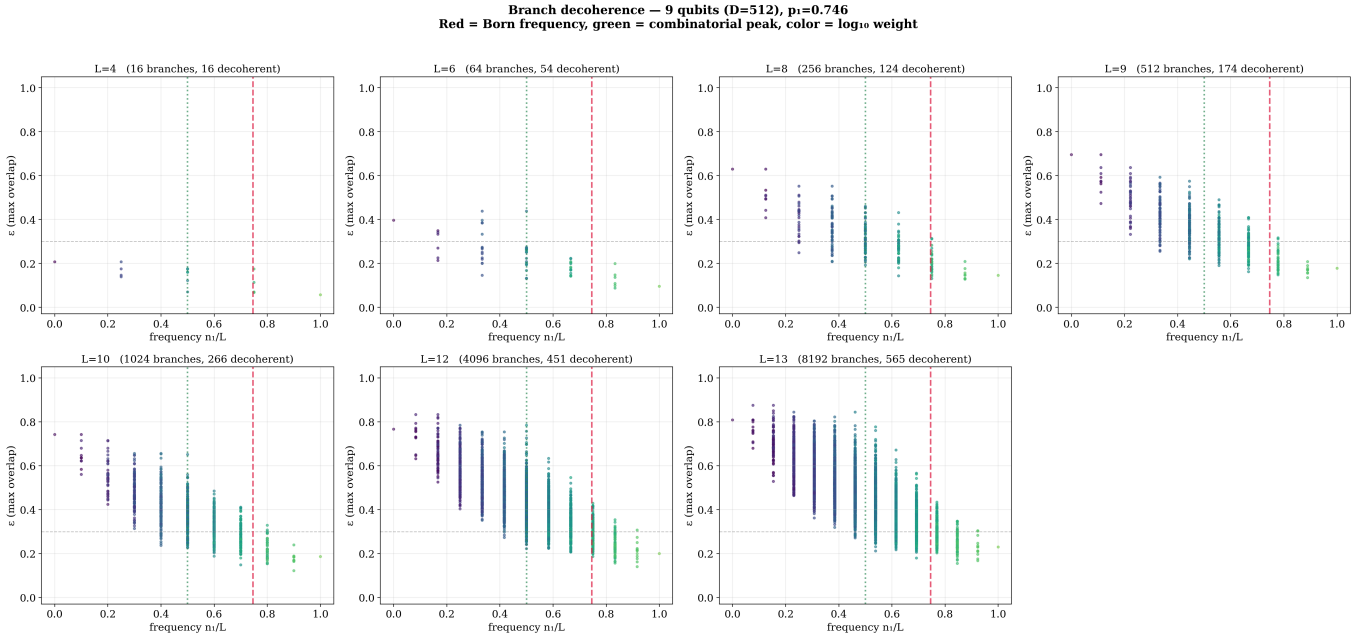

FIG. 2. Recoherence parameter for individual branches at seven values of L. Each dot is one of 2^L branches; color encodes log weight. The decoherence threshold $\varepsilon = 0.3$ is shown as a gray dashed line. At $L = 13$, 8192 branches compete for 512 dimensions; only 565 satisfy $\varepsilon < 0.3$, clustered near the Born frequency (red dashed).

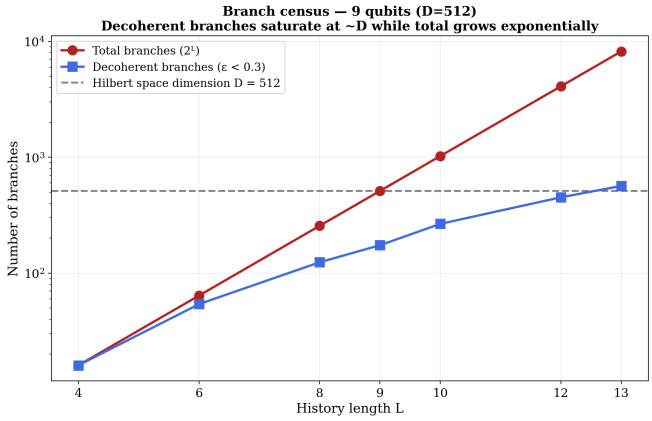

FIG. 3. Branch census: total branches (red) versus decoherent branches with $\varepsilon < 0.3$ (blue). Growth slows dramatically around $D = 512$ (black dashed), while total branches grow as 2^L.

branches—but it sets the scale at which the competition intensifies.

The frequency distribution of survivors matches a binomial with Born probability $p_1 = 0.746$, not the combinatorial distribution from naive branch counting.

E. The All-Zeros and All-Ones Asymmetry

Two extreme branches tell the story most vividly.

The *all-tails* branch—tails on every single flip. This is the most anti-Born history possible. Its weight is 2×10^{-8} (essentially zero), and its $\varepsilon = 0.77$—it overlaps heavily with neighboring branches and fails the approximate decoherence criterion by a wide margin.

The *all-heads* branch—heads every time. There is only one such history out of 4096, so by branch counting it is just as rare as all-tails. But the coin is loaded 75% heads, so this outcome is not absurd. Its weight is 0.025 (a million times larger than all-tails), and its $\varepsilon = 0.20$—it remains approximately decoherent. It has no siblings with the same frequency to compete with, so it retains its share of the Hilbert space.

In this model, anti-Born branches do not just have low weight. They also fail the approximate decoherence criterion used to identify distinct record-bearing histories.

F. Individual Branches

Finally, we pick four individual branches and trace their evolution step by step (Figure 4).

Each row shows one branch's outcome sequence, how its running frequency $n_1(k)/k$ evolves flip by flip, and its weight $\|\psi(\mathbf{x}, k)\|^2$ at each step. The Born-typical branch sees its frequency converge to 75% and retains significant weight. The anti-Born branches drift toward 50% and their weight drops by many orders of magnitude.

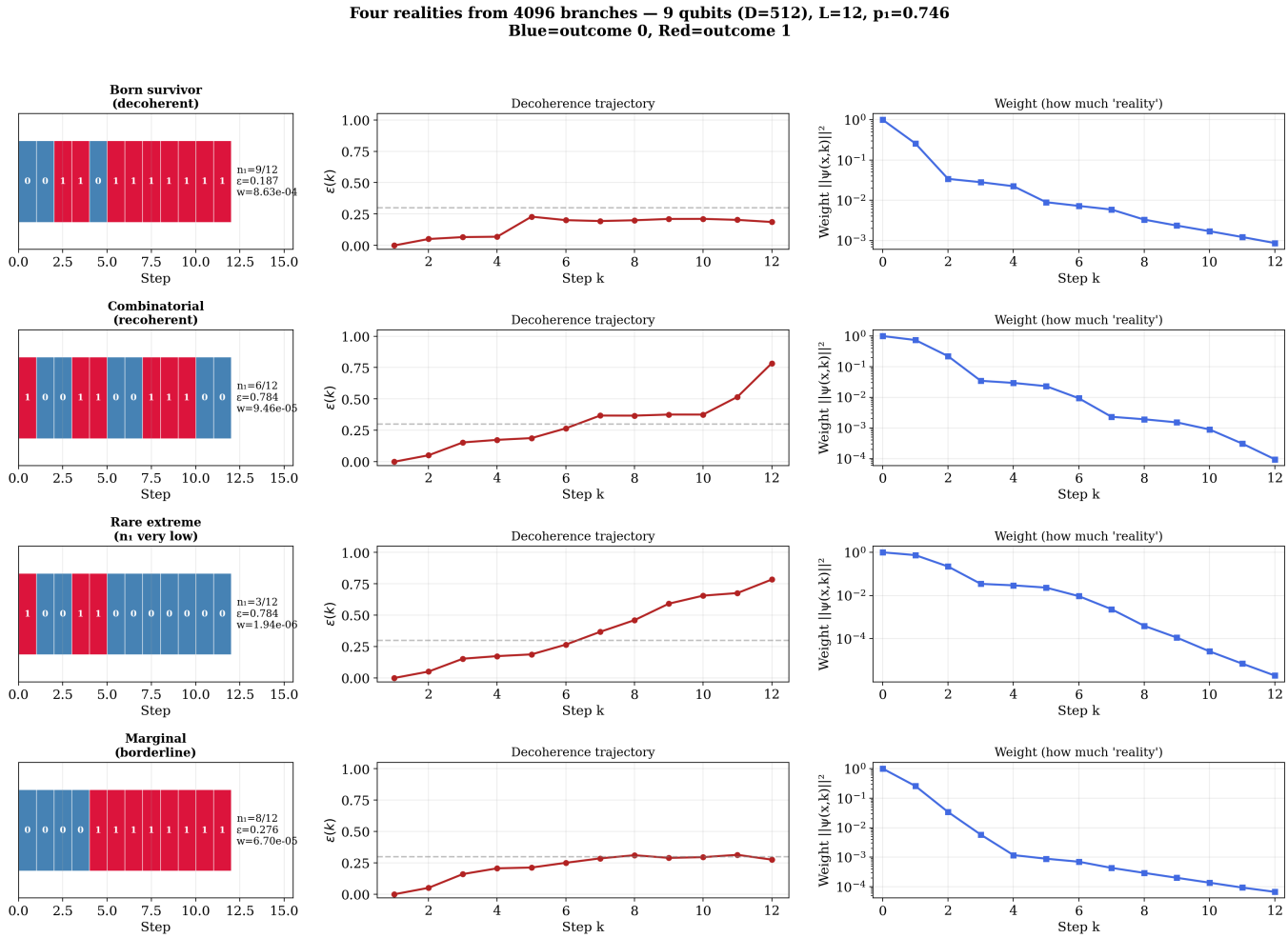

FIG. 4. Four representative branches at $L = 12$. Each row shows: the outcome sequence (blue = 0, red = 1), the running frequency $n_1(k)/k$ converging (or not) to p_1, and the branch weight $\|\psi(\mathbf{x}, k)\|^2$ at each step.

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V. MULTI-SCALE BORN FILTERING

We now ask whether the filtering operates simultaneously at multiple spatial scales of coarse-graining on the same Hilbert space.

We split 8 qubits ($D = 256$) into two groups of 4 (left: qubits 0–3, right: 4–7) and apply independent Hamming-weight projectors (HW ≥ 2) to each group at every time step. For example, if the left group has 3 qubits in $|1\rangle$ and the right group has 1, the joint outcome is $(1, 0)$—left fires, right does not. The four joint outcomes $(00, 01, 10, 11)$ partition $\mathcal{H}$ into orthogonal subspaces. From these we construct two scales of binary projectors:

- **Scale 1**: each group independently ($p_1 = 11/16 = 0.688$),
- **Scale 2**: "at least one group shows 1," i.e. $\Pi_2 = I - \Pi_{0,L} \otimes \Pi_{0,R}$ ($p_1 = 231/256 = 0.902$).

Both are proper orthogonal projectors; the scale-2 Born probability is determined by $\mathrm{rank}(\Pi_2) = 231$, independently of whether the observer knows the spatial decomposition.

Figure 5 shows the frequency-class ε at both scales. Each curve dips near its respective Born frequency ($p_1 = 0.69$ at scale 1, $p_1 = 0.90$ at scale 2), with the larger projector asymmetry at scale 2 producing a stronger gap ($9\times$ vs. $2\times$ at $L = 20$). Figure 6 confirms that the dip develops progressively with L at both scales.

The scale-2 observer perceives $p_1 = 231/256$ as a fundamental property of their projector. The factorization into spatial subgroups—$p_2 = 1 - (1 - p_{1,L})(1 - p_{1,R})$—is invisible at the coarser scale, just as molecular-scale Born probabilities do not reveal the underlying atomic projectors.

A. Emergent Decoherence

The two scales have different branching rates. The fine-grained (scale-1) description has four outcomes per

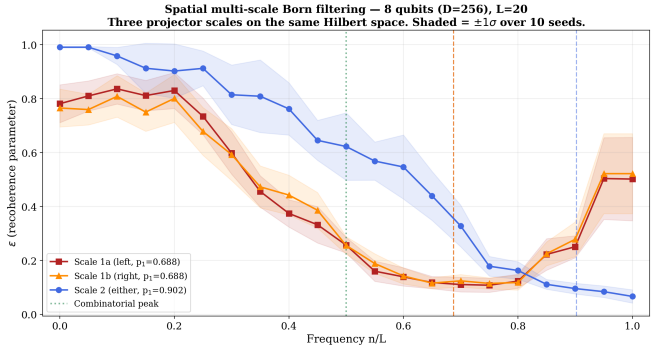

FIG. 5. $\varepsilon(n/L)$ at two spatial scales on an 8-qubit Ising chain ($D = 256$, $L = 20$, 10 Haar-random seeds). Scale 1 curves (red, orange) dip near $p_1 = 0.69$; scale 2 (blue) dips near $p_1 = 0.90$. Shading: $\pm 1\sigma$.

step, producing 4^L branches; the coarse-grained (scale-2) description has two, producing 2^L. Both compete for the same $D = 256$ dimensions, but the finer description exhausts them faster.

We track three nesting metrics across L (averaged over 5–20 seeds, $\varepsilon < 0.3$). S1/S2 denote the mean number of decoherent branches at each scale.

- *Containment*: fraction of scale-1 survivors inside a surviving scale-2 branch.
- *Emergence*: fraction of scale-2 survivors whose scale-1 constituents are *all* dead.
- *Leak*: fraction of dead scale-2 branches containing any scale-1 survivor.

L	4^L	S1	S2	Contain.	Emerg.	Leak
4	256	42	8	98%	45%	4%
5	1 024	25	10	98%	57%	1%
6	4 096	6	10	98%	74%	0%
7	16 384	<1	11	100%	95%	0%
8	65 536	0	11	—	100%	0%

Containment is near-perfect; leakage is near-zero. Decoherence flows from fine to coarse, never the reverse.

Figure 7 makes the structure visible. Each row is one scale-2 branch; the left panel shows its ε (blue = decoherent, red = dead), and the right panel shows every scale-1 branch inside it (blue dots = decoherent, gray = dead). The blue dots appear *only* inside blue-bar rows: no dead scale-2 branch contains a surviving scale-1 branch.

The emergence column is the main result. By $L = 8$, scale 1 is entirely dead: zero out of 65 536 branches satisfy $\varepsilon < 0.3$. Yet ~ 10 scale-2 branches remain decoherent—each a coherent sum of individually dead scale-1 branches. The coarse-grained world supports Born-typical frequencies even though none of its microscopic constituents retain a separate identity.

The mechanism is dimensional: four outcomes per step produce 4^L branches, oversubscribing $D = 256$ by $256\times$ at $L = 8$; two outcomes per step produce 2^L, reaching only $1\times$. The coarser observer creates fewer branches and survives longer—not despite seeing less, but because of it.

VI. DISCUSSION: WHY CLASSICALITY LIVES AT THE TOP

The emergence result inverts a common intuition. We normally think of the macroscopic world as a coarse-grained approximation of a sharper microscopic reality. The nesting table says otherwise: by $L = 8$, the fine-grained description has zero decoherent branches out of 65 536. Every scale-1 history has recohered. Yet ~ 10 scale-2 branches remain decoherent, each a superposition of thousands of individually dead scale-1 histories. In the sense that matters for classical reality (distinct histories, stable records, Born-rule statistics), the macroscopic description is not a degraded version of the microscopic one. It is the level at which distinguishable worlds exist.

The mechanism is dimensional: fine-grained branching produces 4^L histories, exhausting the Hilbert space before coarse-grained branching (2^L) does. The containment and leakage metrics confirm that decoherence flows strictly from fine to coarse, never the reverse. A hypothetical observer tracking physics at full microscopic resolution would find no consistent classical history. Only a coarse-grained observer, one who compresses microstates into macroscopic readings, creates few enough branches to maintain decoherence.

VII. LIMITATIONS

This work has clear limitations.

- **Small system.** We test $D = 128$ to $D = 1024$ (7–10 qubits). Appendix A 1 shows the filtering gap grows with D at matched projector asymmetry, consistent with Strasberg's $D^{-\alpha}$ prediction [9], but the accessible range spans less than one order of magnitude. Establishing the asymptotic scaling requires larger systems.
- **Classical simulation only.** We simulate the full state vector on a classical computer. This limits us to small systems. A quantum computer could in principle probe larger Hilbert spaces, but the analysis requires tracking exponentially many branches, which is inherently hard.
- **Single Hamiltonian family.** We test only the transverse-field Ising chain. The 1D transverse-field Ising model is formally integrable (via Jordan–Wigner), unlike the chaotic random matrix models used by Strasberg. However, at 9 sites with open boundary conditions near the critical point, finite-size effects introduce sufficient dynamical complexity for the filtering to appear. Whether a genuinely

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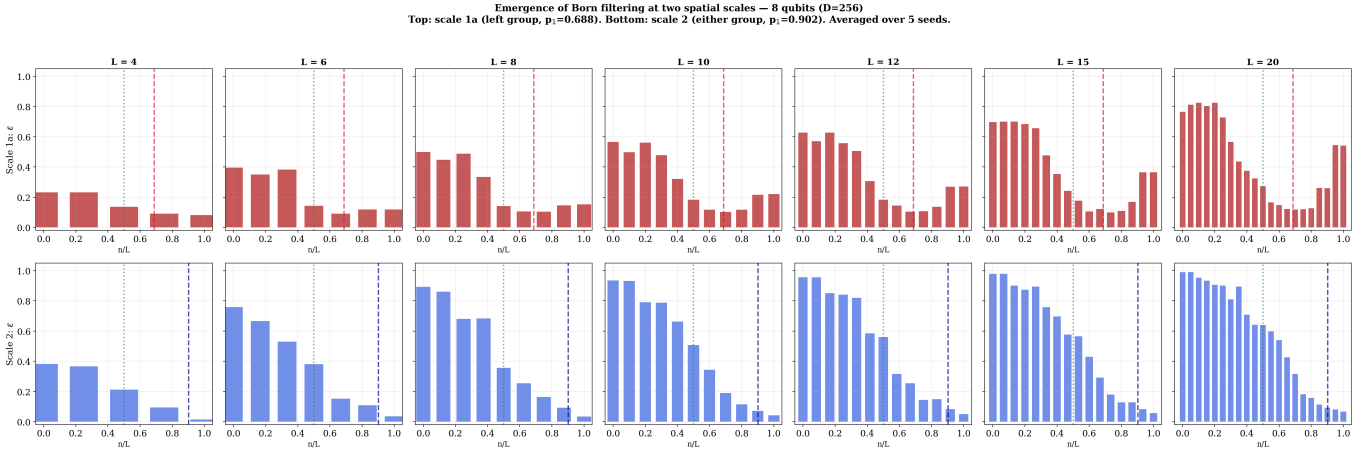

FIG. 6. Emergence of the filtering dip at two spatial scales as L grows. Top: scale 1 (left group, $p_1 = 0.69$). Bottom: scale 2 (either group, $p_1 = 0.90$). Averaged over 5 seeds.

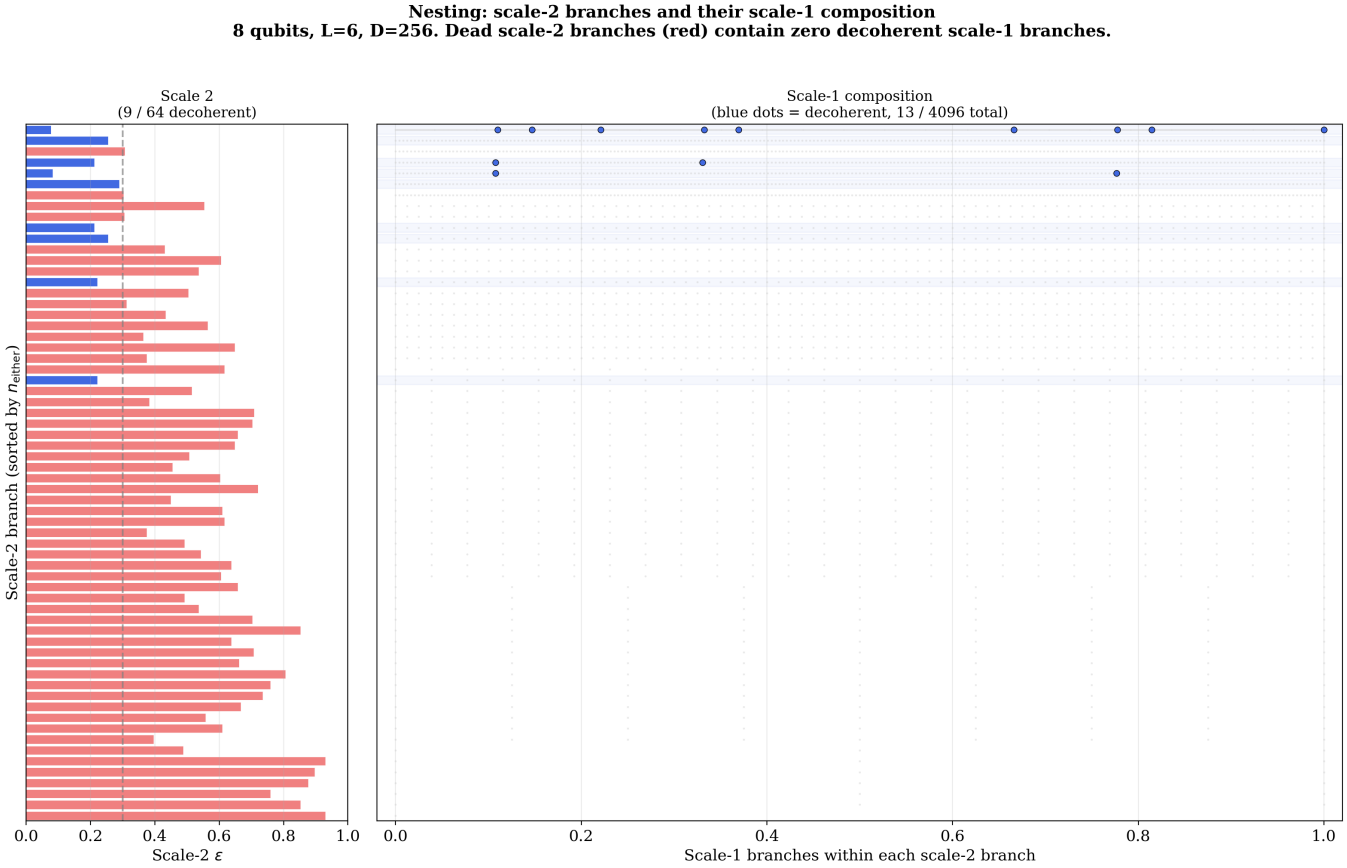

FIG. 7. Nesting at $L = 6$ (4096 scale-1 branches, 64 scale-2 branches, $D = 256$). Left: scale-2 ε per branch (blue = decoherent, red = dead). Right: scale-1 members within each scale-2 branch (blue dots = decoherent). All 13 scale-1 survivors live inside decoherent scale-2 branches; several decoherent scale-2 branches contain no surviving scale-1 members (emergent decoherence).

non-integrable Hamiltonian (e.g., adding a longitudinal field $h_z Z$) would strengthen the effect is an open question and a natural extension of this work.

- **Projector dependence partially explored.** We test four projectors (Appendix A 2): three Hamming-weight thresholds and a spatially biased projector. All asymmetric projectors ($p_1 \neq 0.5$) show filtering, including the spatial projector that is structurally unlike Hamming weight. However,

the filtering strength depends primarily on the projector asymmetry $|p_1 - 1/2|$, not the projector type. When $p_1 = 0.5$, Born and combinatorial peaks coincide and no filtering occurs—as predicted, but limiting the range of independent tests. We have not tested projectors selected by physical symmetries (e.g., gauge invariance).

- **Time-step dependence.** The Born–combinatorial gap depends on Δt, requiring sufficient evolution between projections for the dynamics to spread correlations. The reported $3.1\times$ gap is specific to $\Delta t = 1.2$.

- **No environment.** Real decoherence involves tracing out an environment. Our system has no environment—the 9 qubits are the whole universe. This is intentional (it matches Strasberg's formalism, where the system itself is the recorder), but it is a simplification.

- **Multi-scale analysis limited to two spatial levels.** We test one spatial coarse-graining step (two 4-qubit groups $\rightarrow$ one joint projector) on 8 qubits ($D = 256$). The argument predicts that deeper hierarchies (e.g., groups of groups) should also show Born filtering, but verifying this requires larger systems. The 8-qubit system is smaller than the 9-qubit system used for the single-scale analysis, limiting the filtering strength at scale 1.

VIII. CONCLUSION

In transverse-field Ising chains from 7 to 10 qubits ($D = 128$ to 1024), the recoherence parameter ε dips near the Born frequency, with filtering gaps ranging from $1.9\times$ to $6.1\times$ depending on system size and projector asymmetry. The effect is not specific to a single coarse-graining: it appears for Hamming-weight projectors at multiple thresholds and for a spatially biased projector that examines only a subsystem. The filtering strength scales primarily with the projector asymmetry $|p_1 - 1/2|$ and grows with Hilbert space dimension at fixed asymmetry. A non-scrambling control confirms that projector asymmetry alone is insufficient: scrambling dynamics are required (Appendix A 4). At the individual-branch level, the finite Hilbert space preferentially preserves Born-typical branches while anti-Born branches lose approximate decoherence. This is consistent with Strasberg's analytical results and extends the qualitative picture from random matrices to a local, physically motivated Hamiltonian.

The multi-scale analysis reveals that Born filtering operates simultaneously at multiple spatial scales. When 8 qubits are split into two groups with independent Hamming-weight projectors, the single-group level ($p_1 = 0.69$) and the coarser "at least one group active" level ($p_1 = 0.90$) each show Born filtering independently, with the larger projector asymmetry at the coarser level producing stronger filtering ($9\times$ vs $2\times$ gap). The coarser Born probability derives from the group-level projector ranks, but this decomposition is invisible to the coarser observer.

A natural next step is to test the filtering mechanism in a lattice gauge theory. In a $\mathbb{Z}_2$ or $U(1)$ lattice gauge model, the Hilbert space factorizes into gauge-invariant sectors (charge sectors, flux sectors) that provide a physically dictated projector hierarchy—unlike the Hamming-weight projectors chosen by hand here. The coarse-graining from link variables to plaquette observables to Wilson loops would give a concrete multi-scale structure where each level's projectors are fixed by the gauge symmetry. If Born filtering survives in that setting, the case for a physically grounded mechanism becomes considerably stronger.

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Appendix A: Robustness and Scaling

We present three robustness checks that address natural concerns about the generality of the filtering pattern

Projector	Type	p_1	$\varepsilon_{\text{Born}}$	Gap
Hamming ≥ 3	Global HW	0.910	0.057	10.1×
Hamming ≥ 4	Global HW	0.746	0.087	3.1×
Left-block $\geq 2/5$	Spatial	0.812	0.083	4.4×
Hamming ≥ 5	Global HW	0.500	0.144	1.0×

TABLE I. Born–combinatorial gap for four projectors. "Left-block $\geq 2/5$" examines only the first 5 of 9 qubits (outcome "1" if at least 2 are in state $|1\rangle$). The Hamming ≥ 5 row serves as a null control: $p_1 = 0.5$ means Born and combinatorial peaks coincide.

reported in Sec. IV C: scaling with Hilbert space dimension, dependence on the coarse-graining projector, and sensitivity to the decoherence threshold.

1. Scaling with System Size

We repeat the frequency-class analysis for Ising chains of 7, 8, 9, and 10 qubits ($D = 128$, 256, 512, 1024), varying the Hamming-weight threshold to explore different values of p_1.

Figure 8 shows the ε profile at the largest L tested for each configuration. The qualitative shape—a dip near the Born frequency and a rise toward the combinatorial peak—is present across all system sizes and all asymmetric projectors ($p_1 \neq 0.5$).

The Born–combinatorial gap is governed by two factors (Fig. 9). The dominant factor is the projector asymmetry $|p_1 - 1/2|$: configurations with p_1 close to 0.5 show only marginal filtering (gap $\approx 1.5\times$), while strongly asymmetric projectors ($p_1 > 0.8$) produce gaps exceeding 6×. This is expected: when $p_1 = 0.5$, the Born and combinatorial peaks coincide, leaving nothing to filter. The second factor is system size: at comparable p_1, larger D yields a larger gap. For example, at $p_1 \approx 0.77$–0.83, the gap increases from 2.3× ($D = 128$) to 3.1× ($D = 512$) to 6.1× ($D = 1024$).

2. Projector Independence

We test four projectors on the 9-qubit chain ($D = 512$, $L = 15$):

The left-block projector is structurally different from Hamming weight: it ignores 4 of the 9 qubits entirely and applies a low threshold to the remaining 5. Despite this, it produces a 4.4× gap, confirming that the filtering pattern is not an artifact of the specific Hamming-weight coarse-graining.

3. Threshold Sensitivity

The branch-level analysis in Sec. IV D uses the operational threshold $\varepsilon < 0.3$. Figure 11 shows how the results depend on this choice.

The survivor count is threshold-sensitive (ranging from 19 at $\varepsilon < 0.2$ to 3037 at $\varepsilon < 0.5$), but the qualitative conclusion is not: at every threshold tested, the surviving branches cluster at Born-typical frequencies. The threshold $\varepsilon = 0.3$ sits at the knee of the survivor-count curve, where the count transitions from the tight-threshold regime (few, highly decoherent survivors) to the loose-threshold regime (many survivors with significant mutual overlap).

4. Non-Scrambling Control

We replace the Ising Hamiltonian with a product rotation $U_{\text{prod}} = e^{-i\theta X \otimes m}$ ($\theta = h_x \Delta t$), which generates no entanglement. Projectors, initial states, and p_1 are unchanged. The product rotation produces no stable filtering: the Born–combinatorial gap collapses to 1.3× and $\varepsilon_{\text{Born}} \approx 0.5$ at all L (Figure 12), confirming that scrambling is a necessary ingredient.

DRAFT – April 5, 2026

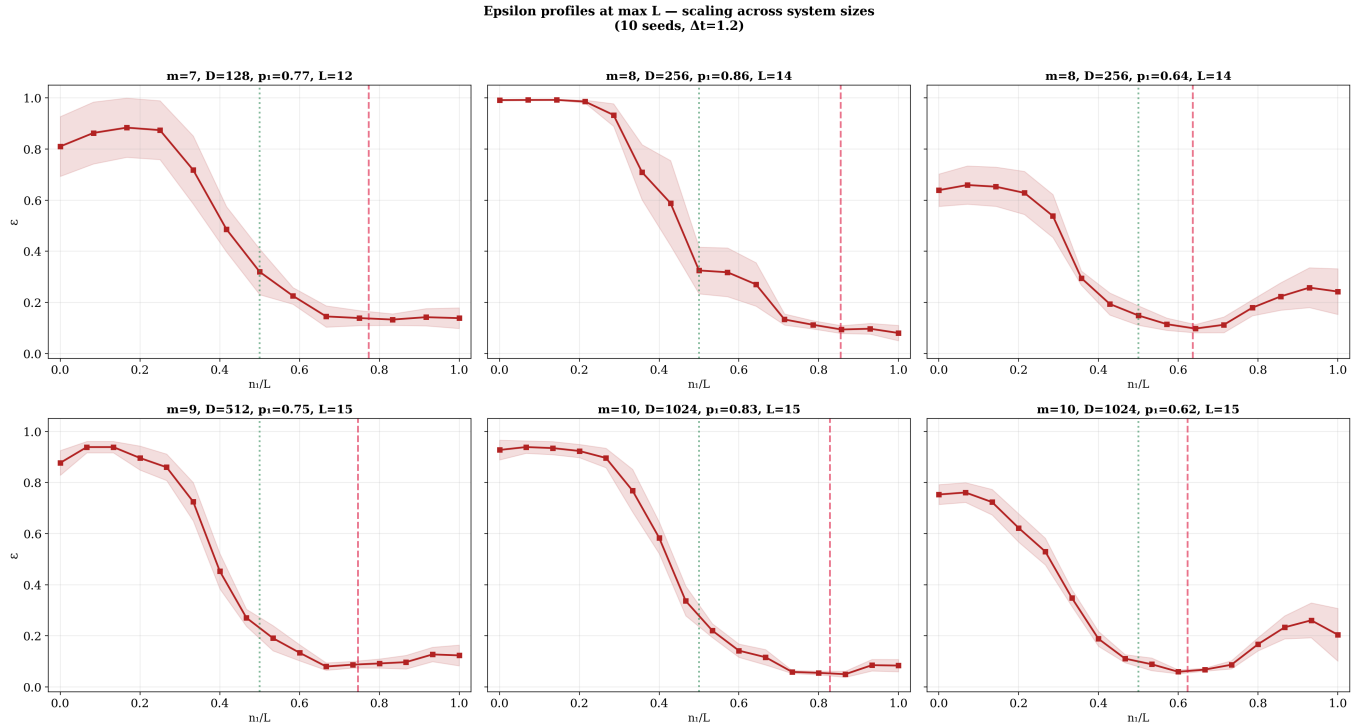

FIG. 8. Recoherence parameter $\varepsilon(n_1/L)$ at the largest L tested for six configurations spanning $D = 128$ to 1024. Red dashed line marks the Born frequency p_1; green dotted line marks the combinatorial peak at 0.5. Each curve is averaged over 10 Haar-random initial states; shading shows $\pm 1\sigma$. The filtering dip near the Born frequency appears at every system size.

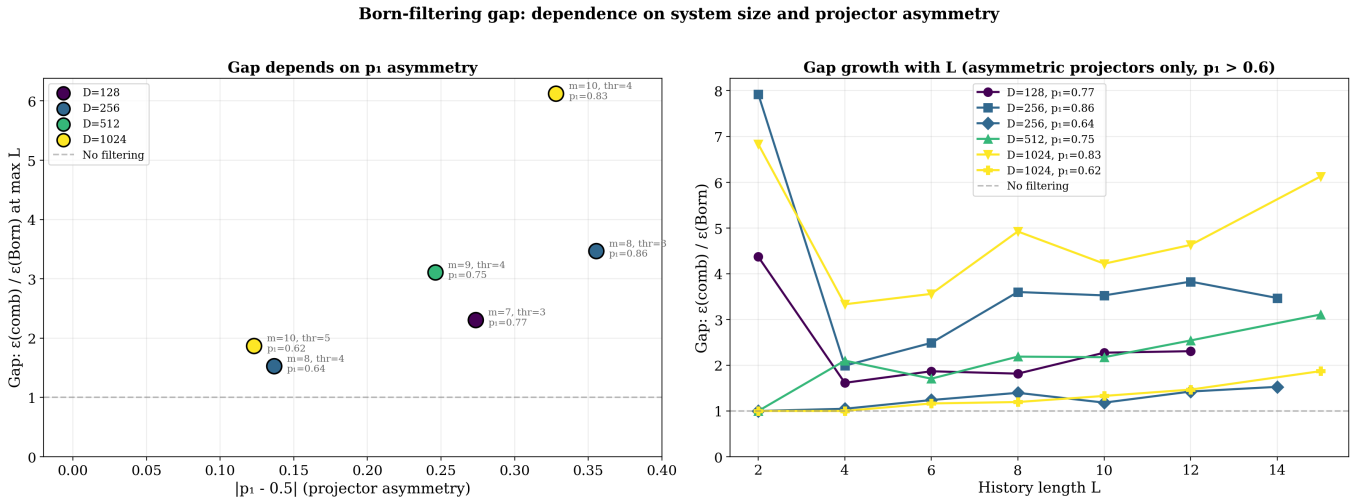

FIG. 9. Left: Born–combinatorial gap at the largest L tested, plotted against projector asymmetry $|p_1 - 1/2|$. Marker color indicates Hilbert space dimension D. At comparable asymmetry, larger D yields a larger gap. Right: gap growth with L for asymmetric projectors only ($p_1 > 0.6$), showing monotonic increase at all system sizes.

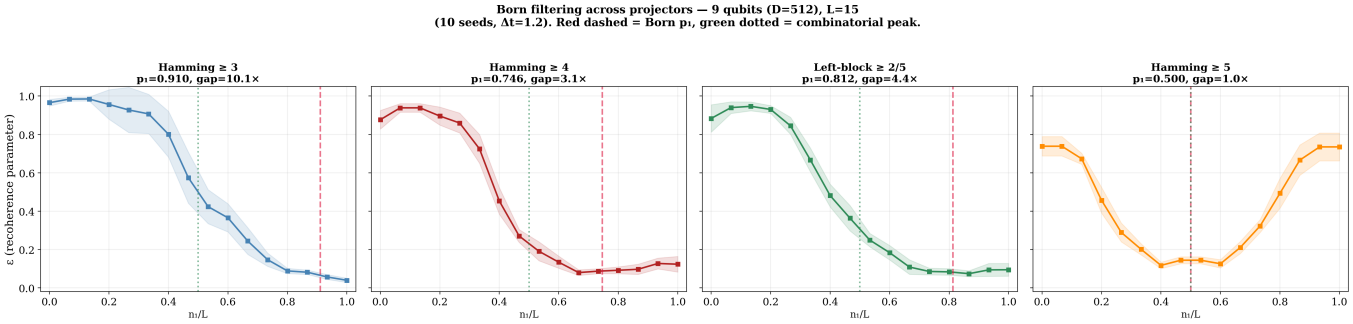

FIG. 10. Recoherence profiles for four projectors on the 9-qubit Ising chain ($L = 15$, 10 seeds). The left-block projector (third panel) examines only a spatial subsystem yet shows the same filtering pattern as global Hamming weight. When $p_1 = 0.5$ (rightmost panel), no filtering occurs.

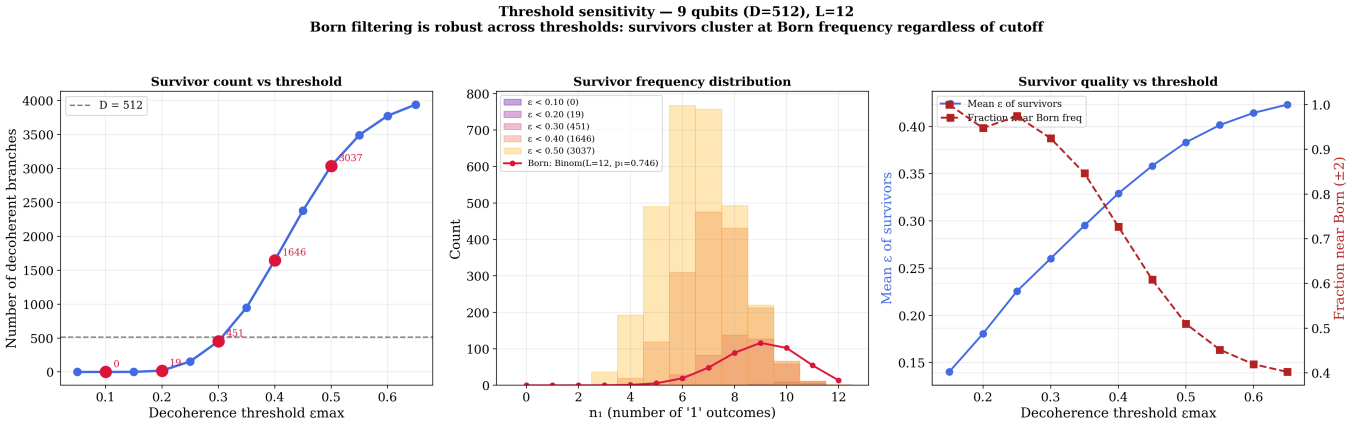

FIG. 11. Threshold sensitivity at $L = 12$ (4096 branches, $D = 512$). Left: number of branches satisfying $\varepsilon < \varepsilon_{\max}$ as a function of the threshold. The curve shows a knee between 0.2 and 0.4. Center: frequency distribution of survivors at five thresholds. Survivors cluster near the Born frequency regardless of threshold. Right: mean ε of survivors and fraction near the Born frequency (± 2) as a function of threshold.

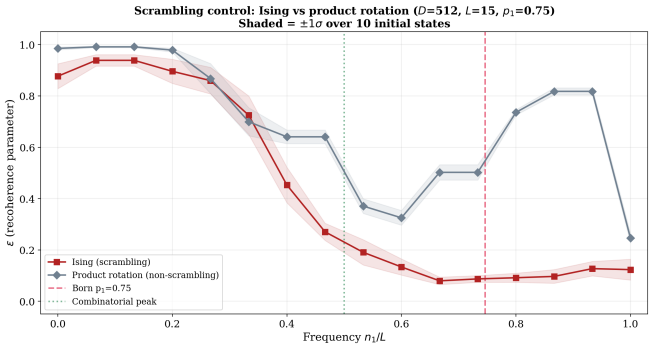

FIG. 12. Non-scrambling control ($D = 512$, $L = 15$, $p_1 = 0.746$, 10 seeds). Ising chain (red) vs. product rotation (gray). The product rotation shows no filtering ($\varepsilon \gtrsim 0.5$ everywhere).